EIN3241 – Human Factors & Ergonomics

Quiz 6 - Dr. Jad Atweh

Exercise 1: Multiple Choice Questions (6 points)

1. In MTM-1, one TMU equals:
   a) 0.036 seconds
   b) 0.36 seconds
   c) 0.0036 seconds
   d) 3.6 seconds

2. The Hawthorne effect refers to:
   a) Temperature affecting worker productivity
   b) Machines breaking during measurements
   c) Analysts becoming biased over time
   d) Workers improving when observed

3. When calculating degrees of freedom, use:
   a) n (sample size)
   b) n - 1
   c) n + 1
   d) 2n

4. A "regular element" in time studies is:
   a) An element that is easy to perform
   b) An element that takes exactly 1 minute
   c) An element that happens every cycle
   d) An element that requires no skill

5. What is the minimum duration for an element in time studies?
   a) 0.01 minutes
   b) 0.02 minutes
   c) 0.04 minutes
   d) 0.10 minutes

6. What are the constant allowances in time studies?
   a) Mental strain and physical fatigue
   b) Setup time and delays
   c) Rest periods and tool changes
   d) Personal needs and basic fatigue

Exercise 2: Time Study (4 points)

An analyst conducted a time study on a packaging operation and collected the following element times:

Cycle	Element Time (sec)
1	12.3
2	11.8
3	12.5
4	12.1
5	11.9
6	12.4
7	12.0
8	11.7

The operator was rated at 105% and the allowance is 12%. Calculate the standard time for this element.

Solution:

Step 1: Calculate average element time (ET)

$ET = \frac{12.3+11.8+12.5+12.1+11.9+12.4+12.0+11.7}{8} = \frac{96.7}{8} = 12.09$ seconds

Step 2: Calculate normal time (NT)

$NT = ET \times \frac{Rating}{100} = 12.09 \times \frac{105}{100} = 12.09 \times 1.05 = 12.69$ seconds

Step 3: Calculate standard time (ST)

$ST = NT \times (1 + \frac{Allowance\%}{100}) = 12.69 \times (1 + \frac{12}{100}) = 12.69 \times 1.12 = 14.21$ seconds

Answer: Standard Time = 14.21 seconds

Exercise 3: Capping a Marker (10 points)

An operator needs to pick up a marker cap from a desk 12” away (the cap is by itself) and place it onto a marker being held in the other hand. Break down this task into its basic motions and calculate the total time in TMU.

My main assumption is that the hands are at the same place originally (no distance between them). If there were distance, it would be give to us...

Motion Description	MTM-1 Code	Time (TMU)
Reach to marker cap 12” (fixed location)	R12B	12.9
Grasp cap (small, by itself)	G1A	2.0
Move cap 12” to marker	M12B	13.4
Position cap on marker (close fit, symmetrical)	P2SE	16.2
Release cap	RL1	2.0
	Total Time:	**46.5 TMU**

Answer: Total time = 46.5 TMU (or 1.674 seconds, or 0.0279 minutes)

Table 13.1 Summary of MTM-1 Data

Table I—Reach—R

Distance Moved (in)	Time (TMU) A	B	C or D	E	Hand in Motion A	B	Case and Description
1/2 or less	2.0	2.0	2.0	2.0	1.6	1.6	A Reach to object in fixed location, or to object in other hand or on which other hand rests.
1	2.5	2.5	3.6	2.4	2.3	2.3	
2	4.0	4.0	5.9	3.8	3.5	2.7	
3	5.3	5.3	7.3	5.3	4.5	3.6	
4	6.1	6.4	8.4	6.8	4.9	4.3	B Reach to single object in location which may vary slightly from cycle to cycle.
5	6.5	7.8	9.4	7.4	5.3	5.0	
6	7.0	8.6	10.1	8.0	5.7	5.7	
7	7.4	9.3	10.8	8.7	6.1	6.5	C Reach to object jumbled with other objects in a group so that search and select occur.
8	7.9	10.1	11.5	9.3	6.5	7.2	
9	8.3	10.8	12.2	9.9	6.9	7.9	
10	8.7	11.5	12.9	10.5	7.3	8.6	D Reach to a very small object or where accurate grasp is required.
12	9.6	12.9	14.2	11.8	8.1	10.1	
14	10.5	14.4	15.6	13.0	8.9	11.5	
16	11.4	15.8	17.0	14.2	9.7	12.9	E Reach to indefinite location to get hand in position for body balance or next motion or out of way.
18	12.3	17.2	18.4	15.5	10.5	14.4	
20	13.1	18.6	19.8	16.7	11.3	15.8	
22	14.0	20.1	21.2	18.0	12.1	17.3	
24	14.9	21.5	22.5	19.2	12.9	18.8	
26	15.8	22.9	23.9	20.4	13.7	20.2	
28	16.7	24.4	25.3	21.7	14.5	21.7	
30	17.5	25.8	26.7	22.9	15.3	23.2	

Table II—Move—M

Distance Moved (in)	Time (TMU) A	B	C	Hand in Motion B	Wt. Allowance Wt. (lb) up to	Factor	Constant (TMU)	Case and Description
1/2 or less	2.0	2.0	2.0	1.7	2.5	0	0	A Move object to other hand or against stop.
1	2.5	2.9	3.4	2.3				
2	3.6	4.6	5.2	2.9	7.5	1.06	2.2	
3	4.9	5.7	6.7	3.6				
4	6.1	6.9	8.0	4.3	12.5	1.11	3.9	
5	7.3	8.0	9.2	5.0				
6	8.1	8.9	10.3	5.7				
7	8.9	9.7	11.1	6.5	17.5	1.17	5.6	
8	9.7	10.6	11.8	7.2				B Move object to approximate or indefinite location.
9	10.5	11.5	12.7	7.9	22.5	1.22	7.4	
10	11.3	12.2	13.5	8.6				
12	12.9	13.4	15.2	10.0	27.5	1.28	9.1	
14	14.4	14.6	16.9	11.4				
16	16.0	15.8	18.7	12.8	32.5	1.33	10.8	
18	17.6	17.0	20.4	14.2				
20	19.2	18.2	22.1	15.6	37.5	1.39	12.5	
22	20.8	19.4	23.8	17.0				
24	22.4	20.6	25.5	18.4	42.5	1.44	14.3	C Move object to exact location.
26	24.0	21.8	27.3	19.8				
28	25.5	23.1	29.0	21.2	47.5	1.50	16.0	
30	27.1	24.3	30.7	22.7				

Table IV—Grasp—G

Case	Time (TMU)	Description
1A	2.0	**Pick Up Grasp**—Small, medium or large object by itself, easily grasped.
1B	3.5	Very small object or object lying close against a flat surface.
1C1	7.3	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter larger than 1/2″.
1C2	8.7	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter 1/4″ to 1/2″.
1C3	10.8	Interference with grasp on bottom and one side of nearly cylindrical object. Diameter less than 1/4″.
2	5.6	**Regrasp.**
3	5.6	**Transfer Grasp.**
4A	7.3	Object jumbled with other objects so search and select occur. Larger than 1″ × 1″ × 1″.
4B	9.1	Object jumbled with other objects so search and select occur. 1/4″ × 1/4″ × 1/8″ to 1″ × 1″ × 1″.
4C	12.9	Object jumbled with other objects so search and select occur. Smaller than 1/4″ × 1/4″ × 1/8″.
5	0	Contact, sliding or hook grasp.

Table V—Position*—P

Class of Fit		Symmetry	Easy to Handle	Difficult to Handle
1—Loose	No pressure required.	S	5.6	11.2
		SS	9.1	14.7
		NS	10.4	16.0
2—Close	Light pressure required.	S	16.2	21.8
		SS	19.7	25.3
		NS	21.0	26.6
3—Exact	Heavy pressure required.	S	43.0	48.6
		SS	46.5	52.1
		NS	47.8	53.4

*Distance moved to engage—1″ or less.

Table VI—Release—RL

Case	Time (TMU)	Description
1	2.0	Normal release performed by opening fingers as independent motion.
2	0	Contact release.

Table 13.1 (continued)

Table VII—Disengage—D

Class of Fit	Easy to Handle	Difficult to Handle
1—Loose—Very slight effort, blends with subsequent move.	4.0	5.7
2—Close—Normal effort, slight recoil.	7.5	11.8
3—Tight—Considerable effort, hand recoils markedly.	22.9	34.7

Table III—Turn and Apply Pressure—T and AP

Weight	Time TMU for Degrees Turned 30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Small—0 to 2 lb	2.8	3.5	4.1	4.8	5.4	6.1	6.8	7.4	8.1	8.7	9.4
Medium—2.1 to 10 lb	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large—10.1 to 35 lb	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

Apply Pressure Case A—10.6 TMU. Apply Pressure Case B—16.2 TMU

Table VIII—Eye Travel Time and Eye Focus—ET and EF

Eye Travel Time $= 15.2 \times \frac{T}{D}$ TMU, with a maximum value of 20 TMU.

where T = the distance between points from and to which the eye travels.
D = the perpendicular distance from the eye to the line of travel T.

Eye Focus Time = 7.3 TMU.